

Md Waqar Moid

Final Year Undergraduate
Department of Biological Sciences and BioEngineering

   waqarmoid23@iitk.ac.in

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2023 - Present	B.Tech	Indian Institute of Technology Kanpur	7.5/10
2023	ISC (XII)	St.Thomas' Boys' School, Kolkata	95.0 %
2021	ICSE (X)	St.Thomas' Boys' School, Kolkata	98.2 %

WORK EXPERIENCE

CarbonTatva AI | *Software Development Intern* | (May '26 - June '26)

- Engineered a **Vectorless RAG chatbot** to accurately query **Bureau of Energy Efficiency (BEE)** guidebooks by replacing vector embeddings with an LLM-driven hierarchical Table of Contents tree to eliminate hallucinations
- Built an offline data pipeline using **PyMuPDF** & **Tesseract OCR** for precise text extraction and TOC-based semantic chunking
- Developed a **FastAPI** backend powered by **Mistral API** with strict verification guardrails and exact page citations
- Deployed the **serverless chatbot** on **Vercel** using local JSON caching, eliminating the need for external vector databases

Unstop | *Content Marketing Intern* (June '25 - July '25)

- Expanded the **central database** of challenging and engaging questions across business sectors, tech, and general trivia
- Researched online **newsletters**, **articles**, and **industry journals** to scout for curated trivia and relevant visual assets
- Authored **1000+** **multiple-choice questions**, enhancing question variety on the platform and boosting **user engagement**

KEY PROJECTS

Interactive Stock Performance Visualisation | *CS661 Course Project* | *Prof. Soumya Dutta* |  (May '25 - July '25)

Objective	Developed an interactive visual analytics dashboard to correlate equity trends with macroeconomic factors
Approach	Processed OHLCV data for 1,700+ NASDAQ stocks; engineered a Greed-Fear Index via RSI & ATR Modeled sector performance overlays against CPI, GDP, & Fed rates ; built US choropleth heatmaps in Plotly
Results	Deployed a multi-page Streamlit platform enabling dynamic cross-sector correlation and risk analytics

Market Driven Brand Repositioning | *MBA631 Course Project* | *Prof. Amit Shukla* (June '25)

Objective	Conducted a 4P market analysis of Spotify to formulate an actionable brand repositioning strategy
Approach	Evaluated competitive positioning in global music streaming; identified key opportunities in social networking Redefined 4Ps (product, price, place, promotion) to pivot platform positioning into a social discovery ecosystem
Results	Delivered a 30-minute presentation and produced a TV Ad commercial detailing strategic repositioning roadmaps

Plant Stress Resistance Classification via Evolutionary Networks | *Prof. Rajesh M. Hegde* |  (May '25 - present)

Objective	Classify plant proteins into 6 abiotic stress categories using sequence data and evolutionary selection pressure
Approach	Extracted amino-acid frequencies and calculated dN/dS ratios to construct a proximal similarity graph Applied small-world network topology to compute graph centrality, clustering, and evolutionary features
Results	Trained an SVM Model achieving 61% accuracy , on par with the benchmark ASRPro model

Stochastic Modelling of Financial Derivatives | *Stamatics, IIT Kanpur* |  (May '25 - July '25)

Objective	Implement stochastic calculus models to accurately price standard European and exotic Asian options
Approach	Built the Black-Scholes-Merton engine in Python and implemented Monte Carlo Brownian simulation Calibrated the Heston stochastic-volatility model against real-world NSE option-chain implied volatilities
Results	Achieved stable numerical convergence and pricing precision across path-dependent and barrier option contracts

BioBytes: Clinical & Drug Discovery Informatics | *BioSoc, IIT Kanpur* |  (May '24 - July '24)

Objective	Develop predictive ML pipelines for hospital readmission risks and drug discovery against SARS-CoV-2
Approach	Executed feature engineering, PCA, and hyperparameter tuning across Random Forest, Logistic Regression, SVM Queried the ChEMBL database and filtered bioactive chemical compounds using Lipinski's Rule of Five
Results	Achieved 78% accuracy and 75% F1-score on readmissions, and identified antiviral drug candidates

Tweet Decoder: Election Sentiment Analysis | *Public Policy & Opinion Cell, IIT Kanpur* (June '24 - July '24)

Objective	Architected an end-to-end NLP intelligence pipeline to evaluate voter polarity and public discourse across 2019 Lok Sabha elections
Approach	- Ingested 100k+ tweets via Twitter API; executed NLTK/spaCy lemmatization, TF-IDF vectorization, and TextBlob/VADER polarity scoring - Implemented Latent Dirichlet Allocation (LDA) via Gibbs sampling to extract latent thematic clusters across political discussions
Results	Quantified macro shifts in voter sentiment, identified key electoral themes, and generated dynamic lexical frequency distributions

ACA253: Full-Stack Web Applications
| ACA, IIT Kanpur
(May '25 - July '25)

Objective	Architect and deploy responsive full-stack web applications with modern frontend frameworks and databases
Approach	- Designed responsive UIs using semantic HTML5, modular CSS3 (Bootstrap), and client-side JavaScript - Built an end-to-end MERN stack (MongoDB, Express, React, Node.js) platform featuring JWT authentication
Results	Deployed an e-commerce platform and a cross-browser compatible interactive online résumé on public cloud

Rare Earth Mineral Value Chains: Risks & Governance
| MBA717M | Prof. Rahul Varman
(Oct '25)

Objective	Synthesize global REE supply data to identify points of failure and formulate strategic policy frameworks
Approach	- Analyzed value chains across US, Australia, China, evaluating economic levers (VAT rebates, limit-pricing) - Modeled geopolitical market failures and designed an allied "Buyers' Club" guaranteed procurement model
Results	Published a comprehensive policy report detailing domestic processing, strategic stockpiles, and recycling

CricEngine
| Self Project |

(July '26 - August '26)

Objective	Build an intuitive, low-latency platform to score and record local cricket matches; a free alternative to paid apps
Approach	- Architected state-driven ball-by-ball scoring logic with structured schemas for overs, wickets, and extras - Implemented client-side state persistence to guarantee zero data loss during live match offline network drops - Designed an intuitive single-tap scorer UX with automated run-rate math, undo actions, and live player metrics
Results	- Integrated secure PIN-based scorer authentication alongside shareable read-only spectator match URLs - Delivered a responsive web app enabling zero-lag ball tracking, instant match summaries, and clean exports

TECHNICAL SKILLS

- Programming Languages:** C, Java, C++, SQL, Python (NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, OpenCV, TensorFlow, PyTorch, TextBlob, Talib, BioPython, VTK)
- Tools/Software:** PostgreSQL, Langchain, Docker, Colab, LATEX, PowerBI, GitHub, Paraview, Tableau, ChromaDB, Vercel

RELEVANT COURSES
(*) Ongoing

Global Value Chains	Development Economics	Project Management*
Big Data Visual Analytics	Marketing Management	Neurobiology
Applied Probability and Statistics	Linear Algebra & Differential Equations	Bioinformatics & Computational Biology
Introduction to Biotechnology	Political Economy of Climate Change	Design for Biosecurity

POSITIONS OF RESPONSIBILITY

Organiser, Inter-IIT Cultural Meet 8.0
(Aug '25 - Dec '25)

- Led a 20+ member team to organise the **Quiz Cup** at **Inter-IIT Cult Meet 8.0**, catering to 4,000+ participants across 23 IITs
- Orchestrated and scheduled **7 flagship quizzes** each featuring **100+ competing teams** and 400+ quizzers
- Handled an allocated budget of **₹3 Lakhs+** and hosted **5 renowned Quizmasters**, managing end-to-end hospitality and logistics

Events Secretary, Media & Cultural Council
(June '24 - April '25)

- Secured **sponsorships** and onboarded multiple **vendor stalls** for flagship events like Treasure Hunts and Cultural Nexus (**Fresher's**)
- Part of organising and conducting core teams for Alfaaz (**Literary Fest**) and Cultural Extravaganza

Secretary, Quiz Club, IIT Kanpur
(April '24 - March '25)

- Represented the institute at premier inter-collegiate competitions, including **Inter-IIT** and **Nihilanth** (Inter-IIT-IIM Quiz Fest)

Academic Mentor, Institute Counselling Service
(July '24 - March '25)

- Conducted one-to-one English **Communication Classes** for first-year UG students who were schooled in vernacular languages

Secretary, BioSoc, IIT Kanpur
(July '24 - March '25)

- Built a comprehensive database of over 100 Biotech and Pharmaceutical companies, actively contacting HRs for campus placements

EXTRA-CURRICULAR ACTIVITIES

- Crowned **National Champion** at **IconIQ Quizverse**, quizzing championship hosted by the **Shiv Nadar Foundation** and secured **₹11 Lakhs+** prize money across multiple competitive stages, outperforming **1,000+ participants** from 430+ colleges across India
- Won **Bronze Medal** in **General Quiz** at **Inter-IIT Cultural Meet 6.0**, competing against **100+ quizzers** from across **23 IITs**
- Won **Bronze Medal** in **Sports Quiz** at **Inter-IIT Cultural Meet 8.0**, competing against **100+ quizzers** from across **23 IITs**
- Secured **3rd** place in **Bithoor Mahotsav Quiz** (UP Ministry of Culture, **₹15,000** prize) and **2nd** in **India Quiz** (UP Tourism)
- Secured **1st** place at the **2022 Kolkata International Quiz Festival** and **Alipore Zoo Festival Quiz** beating 300+ participants
- Secured **1st** position in the **Fresher's Case Study Competition**, organised by the prestigious IITK Consulting Group
- Solved consulting **Problem Statements** as part of Hall 12's team at **Takneek '23 and '24**, IITK's Science & Tech championship
- Led Hall 12 to a **1st** place finish in **Quiz Events** at **Galaxy '26**, IITK's Inter-Hall Cultural Championship
- **Volunteered** for social service and leadership discipline as part of **2 UP CTR** edition of the **National Cadet Corps**